# Yaswanth Muppalla

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***AI Platform Engineer*** | ***Senior Software Engineer*** | *Backend & Agent Systems* | ***AWS/GCP***

## PROFESSIONAL EXPERIENCE

**American Express** | **Senior Software AI Engineer** **Apr 2023 - Present**

**Impact:** Delivered **AWS-based Java/Python APIs** and **IRIS AI platform services** plus **Kafka/Flink/Airflow pipelines on GCP**, improving reporting efficiency by **35%**, achieving **99% data accuracy**, reducing rerun/debug cycles by **3x**, and accelerating **applied AI delivery** for credit-oversight copilots.

- Partnered with **BCA/CCA data pipeline teams** and built JSON-configurable **Apache Flink pipelines on GCP** for ingestion, transformation, joins, validation, and downstream publishing, translating workflow knowledge into **AI-ready pipeline patterns** for credit oversight use cases.
- Built **Airflow workflow automation and API-facing publishing** with dependency controls, rerun handling, **SLA monitoring**, data-quality checks, and failure alerting, creating reliable signals for downstream **AI platform and copilot services**.
- Designed an **IRIS-backed RAG ingestion platform** with document extraction, deduplication, upsert handling, segmentation, **chunking**, metadata enrichment, and batch publishing for **semantic retrieval**.
- Developed a **reusable agent platform harness** combining IRIS semantic search, scoped retrieval, prompt assembly, **source-ledger tracking**, and **grounded response generation** for credit oversight users.
- Partnered with **cross-functional product, data, and AI teams** to engineer an **agent context harness** managing **session state**, selected documents, retrieved evidence, policy context, user intent, and **controlled LLM input assembly**.
- Strengthened **operational excellence, observability, and governance** across the **agent runtime** through retrieval logs, execution traces, citation mapping, **guardrails**, cache behavior, and **audit-friendly response tracking**.

**Skechers** | Software Engineer **Oct 2020 - Mar 2023**

**Impact:** Delivered **15+ Python APIs** and event-driven services, maintained **99.9% uptime**, reduced costs by **22%**, migrated **35M+ records**, and cut incident resolution time by **40%**.

- Developed **15+ Python/Pydantic REST APIs** on microservices-style, event-driven **AWS Lambda/Step Functions**, sustaining **99.9% uptime** and cutting costs **22%**.
- Migrated **35M+ SFMC records** into **MongoDB** and improved query performance by **50%** through indexing, caching, and data-access optimization.
- Integrated **SessionM**, SFMC, MongoDB, and **AWS serverless services** for loyalty/marketing workflows and **sub-500ms transactions**.
- Applied **event-driven API architecture** across Lambda, **Step Functions**, validation, orchestration, and data-access layers.
- Improved **production resilience** with CloudWatch/Datadog alerts, **API failure tracing**, **retry handling**, and SLA-focused incident feedback.

**SOCH Inc.** | Full-Stack Developer - Richmond, VA **Jan 2020 - Sep 2020**

- Built a **Spring Boot** ticketing system with **REST microservices** and full-stack workflow screens for government clients.
- Used **Java ExecutorService/CompletableFuture** for async ticket updates, notifications, and background job handling.
- Managed **concurrency** with synchronized services, **thread-safe collections**, timeouts, and Spring Boot exception handling.

## FOUNDER-LED 0-TO-1 TECHNICAL PROJECT

**OmInsights - Founder-Led Applied AI Platform Build** **Jan 2024 - Present**

**Impact:** Built a **6-component AI platform** with **tenant-scoped APIs**, agent/context orchestration, MCP tool execution, async file processing, and **source-backed intelligence** with tenant isolation, citations, and traceability.

- Architected **Angular UI, FastAPI Business Service, Context Engine, MCP Server**, File Processing worker, and AWS deployment paths with CI/CD, environment promotion, secrets/config handling, and tenant-aware operations.
- Built an **AI evaluation layer** with groundedness, relevance, citation, and policy checks, **LLM-as-a-judge scoring**, and regression tracking.
- Designed **Context Engine flows** for **chat/session memory**, selected context, KB injection, prompt assembly, and **LLM/context harness design** for multi-turn agent orchestration, policy-aware grounding, and session continuity.
- Implemented **MCP/API runtime patterns** and a shared **LLM gateway across platform layers**, centralizing model/provider routing, request policies, retries/fallbacks, tool access, usage logging, and execution traces.
- Created **single-founder execution playbooks** across each platform layer for architecture decisions, implementation, deployment, testing, observability, and recovery, standardizing repeatable delivery and accelerating execution.

## TECHNICAL SKILLS

**AI Platform & Agent Systems: RAG**, embeddings, **vector search**, LangChain/LangGraph, LlamaIndex/LlamaParse, **MCP/FastMCP**, OpenAI APIs, **Amazon Bedrock**, **LLM gateways**, **LLM-as-a-judge evaluation**, **guardrails**, LLMOps, semantic caching, context orchestration

**Backend: Java 17**, **Python**, Scala, **FastAPI**, Pydantic, **Spring Boot**, **REST APIs**, JWT, microservices, Java concurrency, TypeScript/Angular

**Cloud & Data: AWS**, **GCP**, Lambda, ECS/S3/SQS, **Step Functions**, API Gateway/EventBridge, Docker, CI/CD, MongoDB, Redis, MemSQL, Kafka, **Flink/Airflow**

**Retrieval & Storage: OpenSearch Vector DB**, MongoDB, hybrid search, metadata filtering/reranking, chunking, **source ledgers**, searchable publishing, Elasticsearch

**Observability: Datadog**, **CloudWatch**, Kibana, X-Ray, ELK Stack, **structured logging**, SLA alerts, failure tracing

## EDUCATION

**State University of New York, New Paltz** | Master of Science, Computer Science | CGPA: 3.6/4.0 May 2020

**JNTUK** | Bachelor of Technology | CGPA: 8.5/10 2018